Project Objectives

Status: Initial Baseline **Version:** 1.0 **Scope:** Core Deliverables

1. Overall Objective

Design and implement a prototype platform providing a unified view across AWS cloud resources and IoT devices to detect suspicious behaviors, calculate explainable risk, correlate events, and trigger controlled response workflows.

2. Specific Objectives

O1 — Unified Monitoring

Provide a central dashboard displaying cloud findings, device statuses, and response history.

Verification: Demonstrate UI populated with sample resources and findings.

O2 — Cloud & IoT Event Detection

Configure detection for auth failures, API abuses, and abnormal MQTT message volume.

Verification: Execute positive test cases and confirm finding generation.

O3 — Explainable Risk Scoring

Calculate dynamic scores using documented indicators, weights, and severity bands.

Verification: Validate score outputs against pre-calculated unit test cases.

O4 — Incident Correlation

Group multi-source events into unified incidents based on shared entity identifiers and time windows.

Verification: Submit related event pairs and confirm incident aggregation.

O5 — Controlled Automated Response

Trigger automated containment actions (e.g., restrictive IoT policies) for critical thresholds.

Verification: Confirm isolation actions run only on explicitly allowlisted entities.

O6 — Evidence & Auditability

Record comprehensive incident timelines including timestamps, risk factors, and action logs.

Verification: Audit completed incident records for full event trace history.

3. Key Performance Indicators (KPIs)

KPI Indicator	Measurement Method	Initial Target
Detection Accuracy	Expected findings ÷ total test cases	≥ 90% positive cases
Scoring Precision	Score match vs calculated baseline	100% deterministic
Evidence Completeness	Present fields ÷ required audit fields	≥ 95% complete records
Demo Stability	Consecutive successful scenario reruns	3 consecutive runs

4. Primary Deliverables

- System Architecture, Database Schema & Data Dictionary.
- Risk Scoring & Multi-Event Correlation Engine Prototype.
- Automated Containment & Incident Management Dashboard.
- Comprehensive Test Plan, Verification Evidence & Setup Guide.